

POKHARA UNIVERSITY

Level: Bachelor
 Programme: BE
 Course: Probability and Statistics (New)

Semester: Spring
 Year : 2024
 Full Marks : 100
 Pass Marks : 45
 Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) The factories produces two type of car batteries i.e. battery A and battery B. An experiment shows the life of batteries in days which were recorded as follows. 7

Life in days:	500-700	700-900	900-1100	1100-1300	1300-1500
Battery A	5	11	26	10	8
Battery B	4	30	12	8	6

Compare the variability of two type of batteries using coefficient of variation.

- b) State Baye's Theorem. In a certain factory, machines X, Y and Z are all producing springs of same length. Of their production, machine X, Y and Z produces 5%, 4% and 2% defective springs respectively. Of the total production of springs in the factory, machine X produces 25%, machine Y produces 35% and machine Z produces 40%. If one spring is selected at a random from the total springs produced in a day, find: 8

- The probability that it is defective.
- The conditional probability that it was produced by machine Y.

- a) Random variable X has the following probability function: 7

X_i	-2	-1	0	1	2
$P(x)$	0.2	0.1	0.3	0.3	0.1

Find:

- $E(x)$
- $E(2x-3)$
- $V(x)$
- $V(2x-3)$

- b) An office switchboard receives telephone calls at the rate of 3 calls per minute on an average. If receiving of calls follows a Poisson distribution, find the probability of receiving: 8
- No calls in one minute interval.
 - At least 3 calls in a one minute interval.
 - At most 2 calls in 5 minute interval.

OR

In Binomial distribution consisting of 5 independent trials, the probabilities of 1 and 2 successes are 0.4096 and 0.2048 respectively. Find the parameter 'p' of the distribution.

- a) Define rectangular distribution. Derive its mean and variance. 7
- b) In an examination, 10% of the students got less than 20 marks and 5% of the students got over 75 marks. Assuming the distribution to be normal, find the mean standard deviation of the distribution. 8
- a) The joint probability function of random variable X and Y is given by 8

$$f(x, y) = \begin{cases} k(2x + y) & \text{for } 0 \leq x \leq 2, 0 \leq y \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

- Find marginal density of X and Y.
 - Are X and Y independent?
- b) A sample of 20 bulbs, drawn at random from a batch, and discovers that the mean life of the sample bulb is 990 hours with a standard deviation of 22 hours. Find 95% confidence interval for mean. 7
 - a) Explain the criteria of a good estimator. 7
 - b) The score of 10 candidates prior and after training are given below: 8

Prior	84	48	36	37	54	69	83	96	90	65
After	90	58	56	49	62	81	84	86	84	75

Is training effective? (use $\alpha=5\%$)

OR

A machine puts out 16 imperfect articles in a sample of 500. After machine is overhauled, it puts out 3 imperfect articles in a batch of 100. Has the machine improved? (use $\alpha=5\%$)

6. a) In a survey of smoking habits of 100 men and 100 women were asked to classify themselves as smokers or non-smokers, the results summarizes in the table below. 7

Gender Smoking	Men	Women	Total
Yes	54	32	86
No.	46	68	114
Total	100	100	200

Do these data provide any association between the smoking habits and gender? Use $\alpha=5\%$.

- b) The following table gives the ages and blood pressure of 8 women.

8

Age (X):	56	42	36	47	49	42	60	72
Weight (Y):	147	125	118	128	145	140	155	160

- Find the correlation coefficient between X and Y. Also, find the coefficient of determination and interpret it.
- Determine the regression line of Y on X.
- Estimate weight of a woman whose age is 45 years.

Write short notes on: (Any two)

2×5

- Hypergeometric distribution
- Steps in testing hypothesis
- Properties of correlation